

MDS & MyFlexi shop Delivery Pricing API — Technical Integration Guide

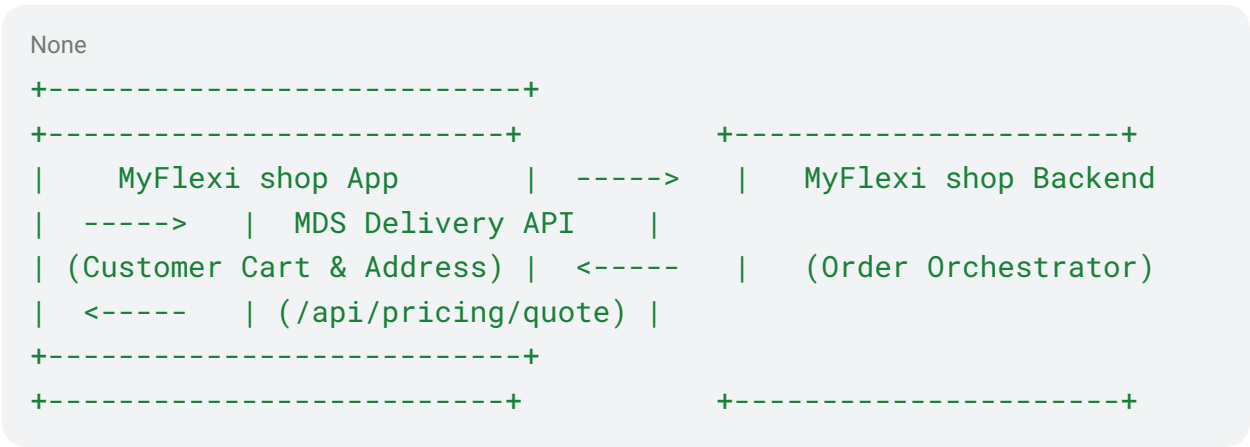
Document Title: Real-Time Delivery Cost Calculation API
Prepared For: MyFlexi shop Engineering & Product Teams
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Document Version: 1.0.0
API Version: v1 (/api/pricing/quote & /api/v1/pricing/quote)
Protocol: HTTPS REST / JSON

1. Overview & Collaboration Architecture

This API is designed specifically for **MyFlexi shop** to seamlessly integrate Mustard Delivery Services (MDS) as the official fulfillment and logistics provider for the MyFlexi shop mobile and web applications.

Whenever a MyFlexi shop user adds items to their shopping cart or proceeds to checkout, the MyFlexi shop backend queries this endpoint to receive:

- Real-Time Cost Price:** The exact delivery cost price to charge or display to the customer.
- Quoted Price Range:** Minimum and maximum price bounds for flexible UI presentation.
- Itemized Transparent Breakdown:** Distance fee, weight handling surcharge, service speed multiplier, goods-in-transit insurance, voucher discounts, and Nigerian statutory VAT (7.5%).
- Estimated Delivery Timelines:** Dynamic delivery windows tailored to the selected service tier.



2. Environments & Base URLs

Environment	Base URL	Description
Sandbox / Staging	https://staging-api.mds.ng	Dedicated environment for MyFlexi shop QA & testing
Production	https://api.mds.ng	High-availability production endpoint
Local Testing	http://localhost:4000	MDS local developer environment

Request Headers

Every request from the MyFlexi shop backend must include:

None

Content-Type: application/json

Accept: application/json

3. Core Pricing Model & Calculation Formula

The pricing engine operates on a **Distance-Primary, Weight-Secondary** model tailored for e-commerce deliveries.

3.1 Primary Variable: Distance Pricing

Distance is the primary pricing driver. The MyFlexi shop backend can provide either:

Direct road distance in kilometers (`distanceKm` or `distance`), **OR**

Pickup (Merchant/Store) and Delivery (Customer) GPS coordinates (`pickup: { lat, lng }, delivery: { lat, lng }`), which automatically computes geodesic distance via the Haversine formula.

Distance Pricing Tiers (Rate Card)

Tier	Distance Range	Baseline Price Range (NGN)	Calculation Logic
Tier 1	1 km – 5 km	Flat ₦2,900	Flat baseline intra-city rate
Tier 2	6 km – 10 km	₦3,000 – ₦6,000	Linear interpolation: $3000 + ((\text{dist} - 5) / 5) * 3000$
Tier 3	11 km – 20 km	₦6,500 – ₦12,000	Linear interpolation: $6500 + ((\text{dist} - 10) / 10) * 5500$
Tier 4	21 km – 30 km	₦13,000 – ₦17,000	Linear interpolation: $13000 + ((\text{dist} - 20) / 10) * 4000$
Tier 5	30 km+	₦40,000+	$40000 + ((\text{dist} - 30) * 200)$ per extra km

3.2 Secondary Variable: Weight Pricing

Weight is evaluated as a secondary alternative:

Standard Light Orders (1 kg – 5 kg): Standard packages (apparel, gadgets, cosmetics, groceries) are covered within the baseline distance fee with zero surcharge (`appliedWeightSurcharge = 0`).

Heavy Orders (> 5 kg): Parcels exceeding 5kg incur a progressive weight handling surcharge added to the distance base fee.

Graceful Fallback: If GPS coordinates or distance cannot be computed, the engine gracefully utilizes the Weight Pricing card and state base zone rate.

Weight Pricing Tiers

Tier	Weight Range	Baseline Price Range (NGN)
Tier 1	1 kg – 5 kg	₦1,700 – ₦12,000
Tier 2	6 kg – 10 kg	₦13,000 – ₦24,000

Tier	Weight Range	Baseline Price Range (NGN)
Tier 3	11 kg – 20 kg	₦ 25,000 – ₦ 48,000
Tier 4	Above 20 kg	₦ 50,000 – ₦ 80,000 (+ ₦ 1,000/kg over 50kg)

3.3 Fulfillment Speeds & Multipliers

MyFlexi shop can offer customer choices across four delivery tiers:

serviceType	Multiplier	Estimated Delivery Window	Description
standard (default)	1.0x	24 – 48 Hours	Standard courier delivery
express	1.5x	2 – 6 Hours	Priority dispatch for urgent orders
same_day	1.8x	Within Business Day	Guaranteed delivery before close of business
cargo	2.0x	24 – 72 Hours	Bulky items / dedicated freight van

3.4 Value-Added Fees, Insurance & Taxes

Goods-in-Transit Insurance (Optional): If `declaredValue` or cart item prices are passed, transit insurance is calculated at 1.5% (0.015) of the total declared value.

Promotional Discounts: If a merchant or campaign `discountCode` is provided, discounts are deducted before tax.

VAT (Value Added Tax): Statutory Nigerian VAT of 7.5% (0.075) is applied to the net subtotal.

Minimum Booking Floor & Rounding: All totals enforce a minimum booking floor of ~~₦~~1,500 and are rounded to the nearest ~~₦~~50 for clean checkout figures.

Master Calculation Formula:

Combined Base = Base Distance Fee + Weight Surcharge
Base Subtotal = Combined Base × Service Multiplier
Subtotal = max(0, Base Subtotal + Insurance Fee - Discount)
Total Cost Price = RoundTo50(max(1500, Subtotal + (Subtotal × 0.075)))

4. API Endpoints Reference

4.1 Calculate Delivery Quote

POST /api/pricing/quote (or /api/v1/pricing/quote)

Calculates the complete delivery cost and returns an itemized breakdown.

Request Schema (application/json)

Field	Type	Required	Default	Description
distanceKm	number	Optional*	—	Direct road distance in kilometers (*Primary variable*).
pickup	object	Optional*	—	{ "lat": number, "lng": number, "address"?: string } Store/Merchant pickup GPS coordinates.
delivery	object	Optional*	—	{ "lat": number, "lng": number, "address"?: string } Customer destination GPS coordinates.
weightKg	number	Optional	1.0	Total order package weight in kilograms (*Secondary variable*).
items	array	Optional	[]	Array of MyFlexi shop cart items [{ name, quantity,

Field	Type	Required	Default	Description
				weightKg, value, category, and itemCode. Example: [{"weightKg": 10, "value": 1000, "category": "Electronics", "itemCode": "ELEC001"}]. Automatically calculates total weight and declared value.
serviceType	string	Optional	"standard"	Speed tier: "standard", "express", "same_day", "cargo".
declaredValue	number	Optional	0	Total declared value of goods in NGN for 1.5% insurance coverage.
originState	string	Optional	—	Nigerian state of dispatch (e.g. "Lagos"). Fallback for zone lookup.
destinationState	string	Optional	—	Destination state (e.g. "Abuja"). Fallback for zone lookup.
discountCode	string	Optional	—	Optional promotional discount code.
pricingStrategy	string	Optional	"distance_primary"	Calculation mode: "distance_primary", "combined", "weight_primary", "distance_only", "weight_only".

** Note: Either distanceKm, pickup/delivery coordinates, or originState/destinationState should be supplied for distance calculation.*

Example Request: MyFlexi shop Cart Checkout

JSON

POST /api/pricing/quote HTTP/1.1

Host: api.mds.ng

Content-Type: application/json

```
{
  "pickup": {
    "lat": 6.5965,
    "lng": 3.3421,
    "address": "MyFlexi Vendor Hub, Ikeja, Lagos"
  },
  "delivery": {
    "lat": 6.4281,
    "lng": 3.4219,
    "address": "12 Admiralty Way, Lekki Phase 1, Lagos"
  },
  "serviceType": "express",
  "items": [
    {
      "name": "Wireless Noise-Cancelling Headphones",
      "quantity": 1,
      "weightKg": 0.8,
      "value": 45000,
      "category": "Electronics"
    },
    {
      "name": "Ergonomic Laptop Stand",
      "quantity": 2,
      "weightKg": 1.2,
      "value": 20000,
      "category": "Accessories"
    }
  ]
}
```

Example Response (200 OK)

JSON

```
{
  "status": "success",
  "data": {
    "quoteId": "QUOTE-MFS-9K2L",
    "currency": "NGN",
    "costPrice": 16900,
    "quotedPriceRange": {
      "min": 10500,
      "max": 19350,
      "recommended": 16900
    },
    "pricingModel": {
      "primaryVariable": "Distance",
      "secondaryVariable": "Weight",
      "calculationStrategy": "distance_primary",
      "distanceSource": "coordinates"
    },
    "breakdown": {
      "distance": {
        "distanceKm": 20.73,
        "tierLabel": "21km - 30km",
        "minFee": 13000,
        "maxFee": 17000,
        "baseDistanceFee": 13292,
        "isPrimary": true
      },
      "weight": {
        "weightKg": 3.2,
        "tierLabel": "1kg - 5kg",
        "baselineWeightFee": 7365,
        "appliedWeightSurcharge": 0,
        "isSecondaryAlternative": true
      },
      "service": {
```

```
    "serviceType": "express",
    "multiplier": 1.5,
    "description": "Express Priority Delivery (2 - 6 hrs)",
    "estimatedHours": {
      "min": 2,
      "max": 6
    }
  },
  "insurance": {
    "declaredValue": 85000,
    "ratePercent": 1.5,
    "fee": 1275
  },
  "discount": null,
  "tax": {
    "vatRatePercent": 7.5,
    "vatAmount": 1177
  },
  "subtotal": 15694,
  "totalCostPrice": 16900
},
"deliveryEstimate": {
  "serviceType": "express",
  "estimatedHours": {
    "min": 2,
    "max": 6
  },
  "description": "Express Priority Delivery (2 - 6 hrs)"
},
"expiresAt": "2026-09-17T10:40:00.000Z"
}
```

4.2 Fetch Rate Card (For Local MyFlexi shop Caching)

GET /api/pricing/rate-card (or /api/pricing/rates)

Returns active distance tiers, weight tiers, tax rates, and service multipliers for local caching on MyFlexi shop servers.

Example Response (200 OK)

JSON

```
{
  "distanceTiers": [
    { "minKm": 1, "maxKm": 5, "minPrice": 2900, "maxPrice": 2900,
      "label": "1km - 5km" },
    { "minKm": 6, "maxKm": 10, "minPrice": 3000, "maxPrice": 6000,
      "label": "6km - 10km" },
    { "minKm": 11, "maxKm": 20, "minPrice": 6500, "maxPrice":
      12000, "label": "11km - 20km" },
    { "minKm": 21, "maxKm": 30, "minPrice": 13000, "maxPrice":
      17000, "label": "21km - 30km" },
    { "minKm": 30, "maxKm": 9999, "minPrice": 40000, "maxPrice":
      40000, "label": "30km and above" }
  ],
  "weightTiers": [
    { "minKg": 1, "maxKg": 5, "minPrice": 1700, "maxPrice": 12000,
      "label": "1kg - 5kg" },
    { "minKg": 6, "maxKg": 10, "minPrice": 13000, "maxPrice":
      24000, "label": "6kg - 10kg" },
    { "minKg": 11, "maxKg": 20, "minPrice": 25000, "maxPrice":
      48000, "label": "11kg - 20kg" },
    { "minKg": 20, "maxKg": 9999, "minPrice": 50000, "maxPrice":
      80000, "label": "Above 20kg" }
  ],
  "vatRate": 0.075,
  "insuranceRate": 0.015,
  "minimumBookingFloor": 1500,
```

```
"serviceMultipliers": {  
  "standard": 1.0,  
  "express": 1.5,  
  "same_day": 1.8,  
  "cargo": 2.0  
}  
}
```

5. Integration Code Samples for MyFlexi shop

5.1 JavaScript / TypeScript (Node.js Axios / NestJS)

TypeScript

```
import axios from 'axios';  
  
interface CartItem {  
  name: string;  
  quantity: number;  
  weightKg: number;  
  value: number;  
}  
  
interface MyFlexiQuoteRequest {  
  pickup: { lat: number; lng: number; address?: string };  
  delivery: { lat: number; lng: number; address?: string };  
  items: CartItem[];  
  serviceType?: 'standard' | 'express' | 'same_day' | 'cargo';  
}  
  
export async function fetchMyFlexiDeliveryQuote(request:  
MyFlexiQuoteRequest) {
```

```

    try {
      const response = await
    axios.post('https://api.mds.ng/api/pricing/quote', request, {
      headers: {
        'Content-Type': 'application/json',
      },
      timeout: 5000,
    });

    const { costPrice, quoteId, breakdown, deliveryEstimate } =
    response.data.data;

    return {
      quoteId,
      deliveryFee: costPrice,
      distanceKm: breakdown.distance.distanceKm,
      estimatedDeliveryTime: deliveryEstimate.description,
    };
  } catch (error: any) {
    console.error('MyFlexi shop pricing quote failed:',
    error.response?.data || error.message);
    throw error;
  }
}

```

5.2 Python (Requests / Django / FastAPI)

Python

```
import requests
```

```
def get_myflexi_delivery_cost(pickup_lat, pickup_lng, dropoff_lat,
dropoff_lng, cart_items, speed="standard"):
```

```

url = "https://api.mds.ng/api/pricing/quote"
headers = {"Content-Type": "application/json"}

payload = {
    "pickup": {"lat": pickup_lat, "lng": pickup_lng},
    "delivery": {"lat": dropoff_lat, "lng": dropoff_lng},
    "items": cart_items,
    "serviceType": speed
}

response = requests.post(url, json=payload, headers=headers,
timeout=5)
response.raise_for_status()
result = response.json()["data"]

return {
    "quote_id": result["quoteId"],
    "cost_price": result["costPrice"],
    "distance_km":
result["breakdown"]["distance"]["distanceKm"],
    "delivery_estimate":
result["deliveryEstimate"]["description"]
}

```

5.3 PHP (Laravel / cURL)

```

PHP
<?php

namespace App\Services;

use Illuminate\Support\Facades\Http;

```

```

class MyFlexiDeliveryService {
    protected $baseUrl = 'https://api.mds.ng/api/pricing/quote';

    public function calculateQuote(array $pickup, array $delivery,
    array $items, string $serviceType = 'standard') {
        $response = Http::timeout(5)->post($this->baseUrl, [
            'pickup' => $pickup,
            'delivery' => $delivery,
            'items' => $items,
            'serviceType' => $serviceType,
        ]);

        if ($response->successful()) {
            return $response->json('data');
        }

        throw new \Exception('Delivery pricing error: ' .
        $response->body());
    }
}

```

6. HTTP Status Codes & Error Handling

HTTP Code	Error Type	Description
200 OK	Success	Quote successfully computed.
400 Bad Request	Validation Error	Coordinates out of range, negative weights, or malformed JSON. Check response body for field validation issues.
404 Not Found	Resource Not Found	Invalid endpoint path or unlisted courier type ID.

HTTP Code	Error Type	Description
429 Too Many Requests	Rate Limit Exceeded	Rate limit reached. Implement retry with backoff.
500 Internal Server Error	Server Error	An internal MDS engine error occurred.

Error Response Format

```
JSON
{
  "statusCode": 400,
  "message": [
    "delivery.lat must be a number conforming to the specified constraints",
    "distanceKm must not be less than 0"
  ],
  "error": "Bad Request"
}
```

7. Recommended Integration Flow for MyFlexi shop Checkout

Checkout Address Step: When a customer selects or enters their shipping address in the MyFlexi shop app, capture destination coordinates.

Fetch Quote: Send order cart items and coordinates to `POST /api/pricing/quote`.

Display Options: Present the calculated `costPrice` on the order summary or display speed options (Standard vs Express).

Order Confirmation: Save the returned `quoteId` with the MyFlexi shop order record.

When initiating dispatch with MDS, reference the quote ID to ensure 100% price parity.

Quote Validity Window: Quotes are valid for **30 minutes** (`expiresAt`). If checkout takes longer, simply re-fetch the quote before payment capture.

8. Technical Support & Point of Contact

MDS Technical Support: api-support@mds.ng

Interactive Swagger Docs: <https://api.mds.ng/api/docs>

Dedicated MyFlexi shop Channel: Available on Slack / WhatsApp partner group.